Multi-Agent Modeling of Human Traffic Dynamics for Rapid Response to Public Emergency in Spatial Networks

Shi, Lee, Yang — IEEE CASE 2024

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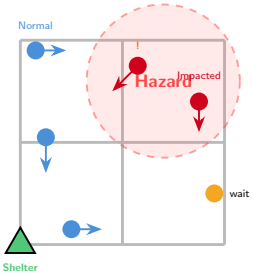
Problem: Simulating Emergency Evacuation

Scenario: A hazard (fire, explosion, flood) strikes a community.
Thousands of pedestrians must evacuate simultaneously.

What makes this hard to simulate?

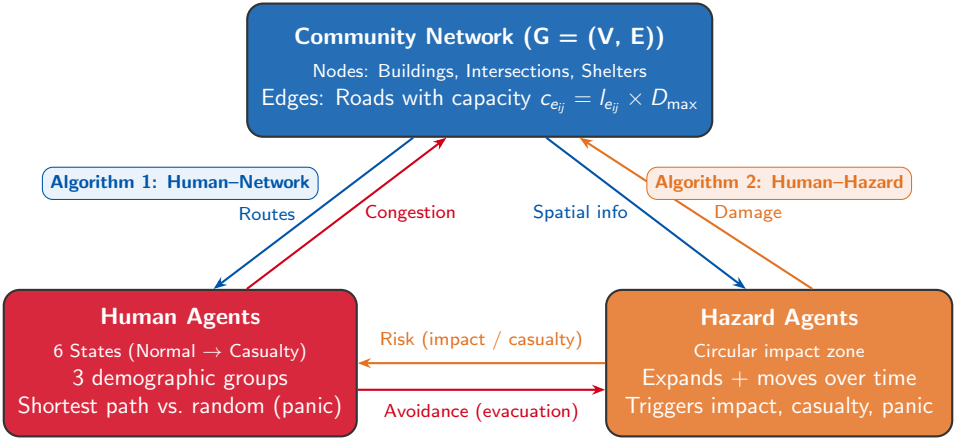
- Each person decides **independently** — where to go, which route to take
- Decisions depend on **local information**: is this road congested? Am I near the hazard?
- **Panic** causes irrational behavior — random routing instead of shortest path
- Congestion is **emergent** — it arises from many individual decisions, not a global rule



Each dot = one agent making its own decision

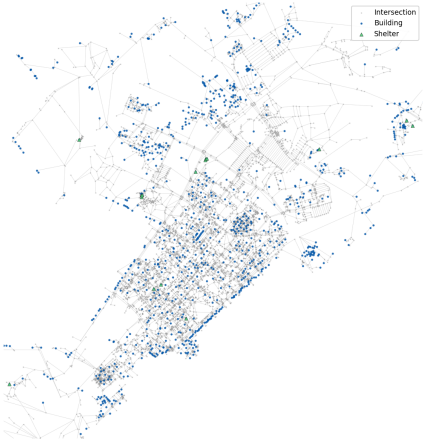
A traditional process-flow model cannot capture this.
⇒ We need **Agent-Based Simulation**: model each pedestrian as an autonomous agent with its own state, position, speed, and decision logic.

Model Overview — Three Components



Shi et al. (2024), Section III

Spatial Network — What Does It Represent?



PSU University Park — from OpenStreetMap

The network is a graph $G = (V, E)$ that represents the **physical environment** where evacuation takes place.

Nodes (V) — Places

- **Building nodes** — where people start (origins) and go (destinations)
- **Intersection nodes** — routing decision points
- **Shelter nodes** — safe destinations during emergency

Edges (E) — Roads

Each edge has two key properties:

- **Length** $l_{e_{ij}}$ — physical distance (meters)
- **Capacity** $c_{e_{ij}} = l_{e_{ij}} \times D_{\max}$ — max pedestrians the road can hold

When flow $\geq$ capacity $\Rightarrow$ **congestion** $\Rightarrow$ agents must queue or reroute.

$D_{\max}$ varies by road type: footway (3), pedestrian zone (8), residential (5) ped/m

Shi et al. (2024), Section III-A

Human Agents — Parameters & Their Roles

What defines each agent?

Core Parameters

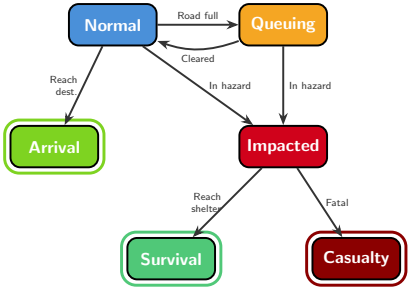
- **Position** — current node or edge — "where am I?"
- **Destination** — building (normal) or shelter (after impact)
- **Speed** — how fast they walk (72–96 m/min by age)
- **State** — behavioral mode — 6 possible states (right)
- **Panic rate** ε_p — probability of irrational behavior

What does panic do?

Normal agent: computes shortest path to destination each step
Panicked agent: picks a random edge at every node

$\Rightarrow$ Higher ε_p = more agents lost = fewer reach shelter

State Transitions



Key insight: Once Impacted, the agent's destination changes from their original building to the nearest shelter. Panic is **irreversible** — once panicked, the agent never recovers rational routing.

Shi et al. (2024), Section III-C, Tables II–III

Hazard Agents — How Danger Spreads

What defines each hazard?

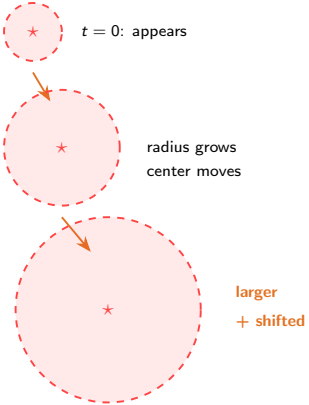
Core Parameters

- **Center** — starting location on the network
- **Radius** $r(t)$ — how far the danger reaches — grows over time
- **Expansion speed** — how fast the radius grows each step
- **Movement speed** — how fast the center moves
- **Lifespan** — how long the hazard remains active
- **Direction** — which way the center drifts

What happens to agents inside the zone?

1. State → **Impacted**, reroute to nearest shelter
2. Each step: probability of **casualty** (higher near center)
3. Each step: probability of **panic** (irreversible)

Lifecycle: emergence → expansion → movement



Multiple hazards ($H_{\max}$) can occur during one simulation.

Shi et al. (2024), Section III-B

How It Works — One Agent's Perspective

Each time step (= 1 minute), **every agent** goes through two checks:

Step 1: Am I in danger? (Algo. 2)

1. **Check**: is my position inside any hazard zone?
2. If **yes** (first time):
   - State → Impacted
   - New destination → nearest shelter
3. If inside zone (every step):
   - Roll for casualty (prob. ↑ near center)
   - Roll for panic (prob. $= \varepsilon_p \times$ group rate)
4. If **no**: skip, proceed to Step 2

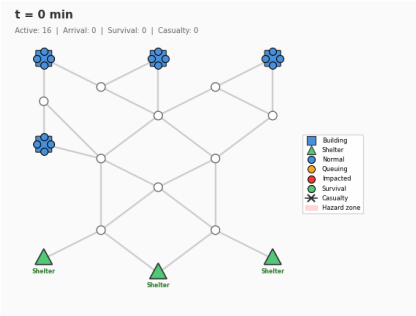
Step 2: Where do I go? (Algo. 1)

1. **Check**: am I at my destination?
   - Yes → Arrival or Survival
2. **Check**: is the road ahead full?
   - All edges full → Queuing (wait)
3. **Decide route**:
   - **Not panicked**: recompute shortest path (A*)
   - **Panicked**: pick a random edge
4. **Move** along chosen edge at my speed

Simultaneous update: all agents decide based on the *same* snapshot of the world, then all positions update at once — no agent has an unfair advantage.

Shi et al. (2024), Algorithms 1–2

Toy Example — Simulation in Action



Timeline

$t = 0$: Normal operation

10 agents depart from **buildings** toward random destinations.

$t = 4$: Hazard emerges

Agents in the zone → **Impacted**, reroute to **shelter**. Some become **casualties** (×).

$t = 10$–25: Evacuation under pressure

- Zone expands → more agents impacted
- **Queuing** at congested edges
- **Panicked** agents (!) walk randomly
- Others follow shortest path to shelter

$t = 25$: Evacuation complete

All agents resolved: Arrival, Survival, or Casualty. Hazard zone still active but no one left inside.

Current Status & Project Direction

What we have done

- ✓ Paper analysis & model understanding
- ✓ Base implementation complete
  (network, agents, hazards, Algorithms 1–2)
- ✓ Toy simulator with all 6 states, congestion, panic, simultaneous update
- ○ Validation against paper: **pending**

Reproduction targets

- **Table V:** Verify network structure (Buildings, Non-building nodes, Edges) for 5 communities:
  PSU Univ. Park, UVA Charlottesville, Virginia Tech, Reading PA, King of Prussia PA
- **Fig. 5:** Impacted Rate (RI) vs. max pedestrian flow ($P_{\max}$), grouped by number of hazard events ($H_{\max}$)
- **Fig. 6:** Survival / Casualty / Leftover Rates vs. Panic Rate (ε_p)

45 parameter combinations ($3 \times 3 \times 5$), multi-seed averaging

Reproduction approach

- Build identical walk networks from OpenStreetMap for each community
- Run full factorial experiment:
  $P_{\max} \in \{2000, 5000, 8000\}$
  $H_{\max} \in \{5, 10, 15\}$ (number of hazard events)
  $\varepsilon_p \in \{10\%, 30\%, 50\%, 70\%, 90\%\}$
- Compare quantitatively with the paper's published Table V, Fig. 5, and Fig. 6

Future work (from the paper's discussion)

- **Additional agent types:**
  Add firefighters, trained volunteers to model organized response alongside evacuation
- **Social force models:**
  More realistic panic behavior using physics-based crowd dynamics (attraction/repulsion)
- **Markov Decision Processes:**
  Probabilistic modeling of irrational decisions triggered by evolving hazard conditions